

AI/ML INTERNSHIP – 4-WEEK PRACTICAL LEARNING TRACK

Week 1 – Task 1

Python Fundamentals

Execution Report with Terminal Screenshots

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1 Introduction

This report documents the solutions to Task 1 of Week 1 (Python fundamentals). The work is split into six scripts, one per core topic. Each script was executed in the VS Code integrated terminal and the output was captured as evidence.

#	Script	Topics covered
1	01_variables_datatypes.py	Types, casting, strings, collections, comprehensions
2	02_control_flow_loops.py	Conditionals, <code>for/while</code> , nested loops
3	03_functions.py	Default args, <code>*args/**kwargs</code> , recursion, lambda
4	04_oop.py	Classes, inheritance, encapsulation, polymorphism
5	05_file_handling.py	Text, CSV and JSON files
6	06_exceptions.py	<code>try/except/else/finally</code> , custom exceptions

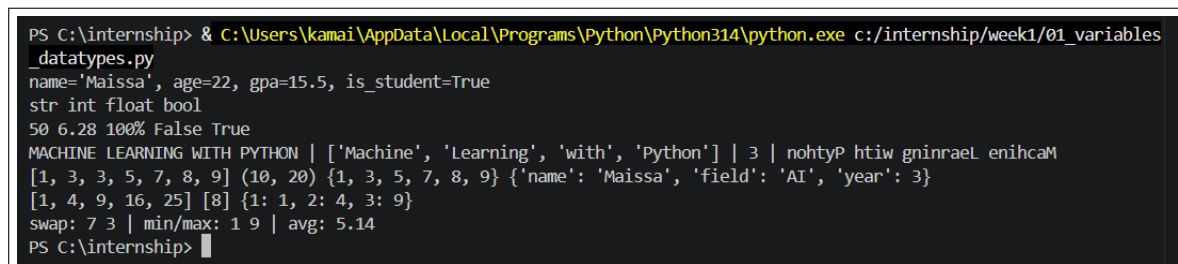
Table 1: Overview of the exercises.

2 Environment

- **Language:** Python 3.14 (standard library only, no external packages)
- **Editor:** Visual Studio Code with the Python extension
- **Operating system:** Windows, PowerShell terminal
- **Working folder:** C:\internship\week1

3 Exercise 1 – Variables and Data Types

Goal: declare variables of the basic types, convert between them, manipulate strings, and use lists, tuples, sets and dictionaries.



```

PS C:\internship> & C:\Users\kamai\AppData\Local\Programs\Python\Python314\python.exe c:/internship/week1/01_variables_datatypes.py
name='Maissa', age=22, gpa=15.5, is_student=True
str int float bool
50 6.28 100% False True
MACHINE LEARNING WITH PYTHON | ['Machine', 'Learning', 'with', 'Python'] | 3 | nohtyP htiw gnirraeL enihcaM
[1, 3, 3, 5, 7, 8, 9] (10, 20) {1, 3, 5, 7, 8, 9} {'name': 'Maissa', 'field': 'AI', 'year': 3}
[1, 4, 9, 16, 25] [8] {1: 1, 2: 4, 3: 9}
swap: 7 3 | min/max: 1 9 | avg: 5.14
PS C:\internship>

```

Figure 1: Output of 01_variables_datatypes.py.

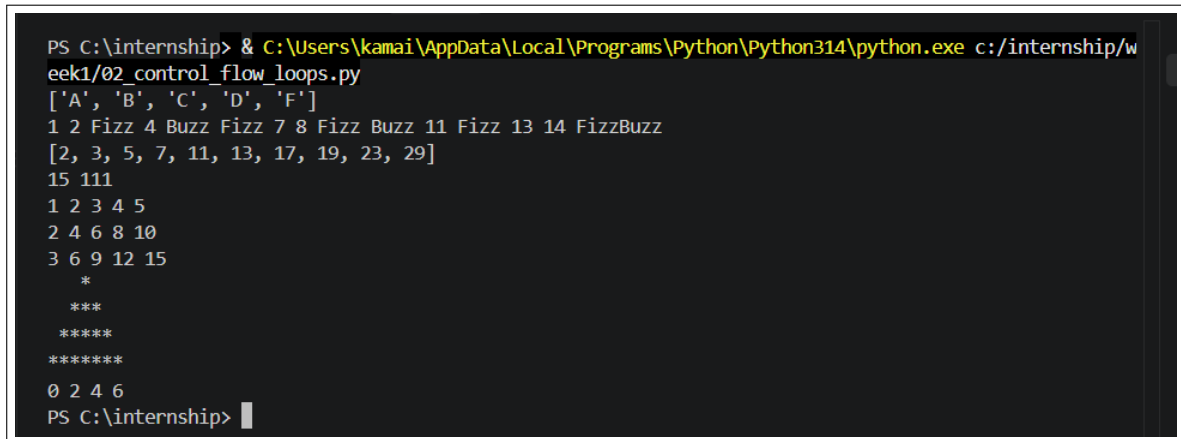
Observations (Figure 1):

- The four base types (`str`, `int`, `float`, `bool`) are printed correctly.
- Casting works as expected: `int("42")+8` gives 50, and `bool(0)` is `False` while `bool("text")` is `True`.

- The set removes the duplicate value 3 from the list, and the dictionary value `year` was updated from 2 to 3.
- List and dictionary comprehensions produce the squares and the `{x: x^2}` mapping.

4 Exercise 2 – Control Flow and Loops

Goal: apply `if/elif/else`, `for` and `while` loops, `break/continue`, and nested loops.



```
PS C:\internship> & C:\Users\kamai\AppData\Local\Programs\Python\Python314\python.exe c:/internship/w
eek1/02_control_flow_loops.py
['A', 'B', 'C', 'D', 'F']
1 2 Fizz 4 Buzz Fizz 7 8 Fizz Buzz 11 Fizz 13 14 FizzBuzz
[2, 3, 5, 7, 11, 13, 17, 19, 23, 29]
15 111
1 2 3 4 5
2 4 6 8 10
3 6 9 12 15
*
***
*****
*****
0 2 4 6
PS C:\internship>
```

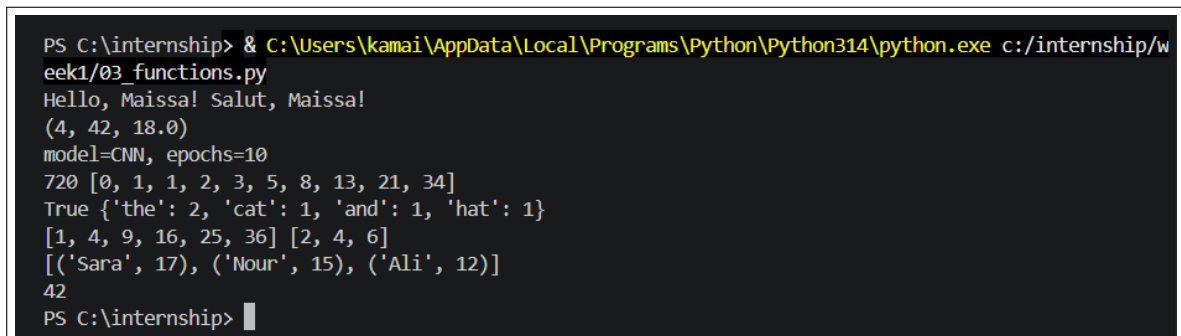
Figure 2: Output of `02_control_flow_loops.py`.

Observations (Figure 2):

- The grading function maps scores 95, 83, 71, 55, 20 to A, B, C, D, F.
- FizzBuzz is correct for 1–15, and the prime list up to 30 is `[2, 3, 5, ..., 29]`.
- The digit sum of 12345 is 15, and the Collatz sequence starting at 27 takes 111 steps.
- Nested loops print a multiplication table and a centred triangle; the `continue/break` demo prints only 0, 2, 4, 6.

5 Exercise 3 – Functions

Goal: write reusable functions with default and variable-length arguments, recursion, lambdas and closures.



```
PS C:\internship> & C:\Users\kamai\AppData\Local\Programs\Python\Python314\python.exe c:/internship/w
eek1/03_functions.py
Hello, Maissa! Salut, Maissa!
(4, 42, 18.0)
model=CNN, epochs=10
720 [0, 1, 1, 2, 3, 5, 8, 13, 21, 34]
True {'the': 2, 'cat': 1, 'and': 1, 'hat': 1}
[1, 4, 9, 16, 25, 36] [2, 4, 6]
[('Sara', 17), ('Nour', 15), ('Ali', 12)]
42
PS C:\internship>
```

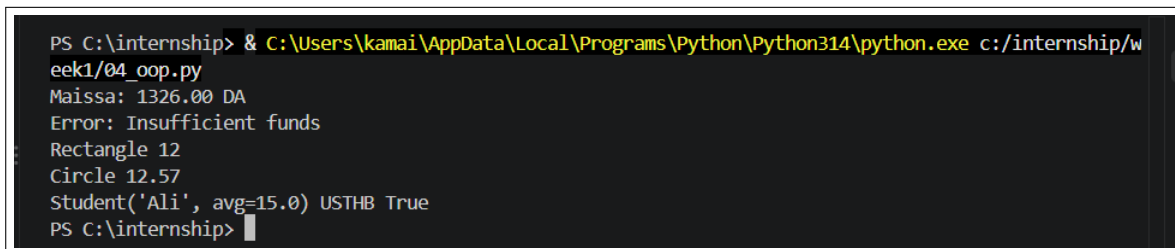
Figure 3: Output of `03_functions.py`.

Observations (Figure 3):

- The default argument produces “Hello” unless a greeting is passed explicitly.
- `stats(*numbers)` returns `(min, max, mean) = (4, 42, 18.0)` for six values.
- `factorial(6) = 720` (recursion) and the first ten Fibonacci numbers are correct.
- `map`, `filter` and `sorted(key=lambda ...)` work as expected, and the closure `make_multiplier(2)` doubles 21 to 42.

6 Exercise 4 – Object-Oriented Programming

Goal: model real objects with classes, inheritance, encapsulation, polymorphism and special methods.



```
PS C:\internship> & C:\Users\kamai\AppData\Local\Programs\Python\Python314\python.exe c:/internship/w
eek1/04_oop.py
Maissa: 1326.00 DA
Error: Insufficient funds
Rectangle 12
Circle 12.57
Student('Ali', avg=15.0) USTHB True
PS C:\internship>
```

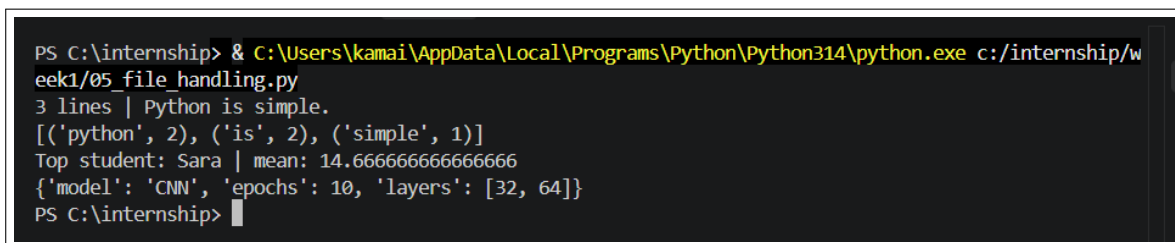
Figure 4: Output of 04_oop.py.

Observations (Figure 4):

- `SavingsAccount` inherits from `BankAccount`: $1000 + 500 - 200 = 1300$, then 2% interest gives 1326.00.
- Withdrawing more than the balance raises a `ValueError`, caught and printed as “Insufficient funds”.
- Polymorphism: `Rectangle` and `Circle` each implement their own `area()` (12 and 12.57).
- `__repr__` and `__eq__` are customised on `Student`, and the class attribute `school` is shared by all instances.

7 Exercise 5 – File Handling

Goal: read and write text, CSV and JSON files using context managers (`with`).



```
PS C:\internship> & C:\Users\kamai\AppData\Local\Programs\Python\Python314\python.exe c:/internship/w
eek1/05_file_handling.py
3 lines | Python is simple.
[('python', 2), ('is', 2), ('simple', 1)]
Top student: Sara | mean: 14.666666666666666
{'model': 'CNN', 'epochs': 10, 'layers': [32, 64]}
PS C:\internship>
```

Figure 5: Output of 05_file_handling.py.

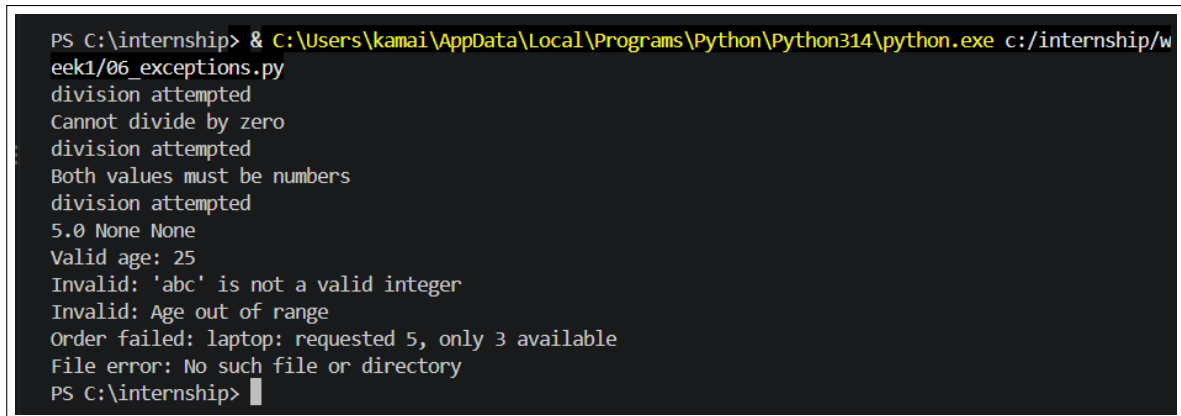
Observations (Figure 5):

- A text file is written, appended to, and read back (3 lines).
- Word frequency with `Counter` shows “python” and “is” each appear twice.

- The CSV round-trip identifies the top student (Sara) and the mean score (≈ 14.67).
- The JSON configuration is saved and reloaded unchanged.

8 Exercise 6 – Exception Handling

Goal: handle errors gracefully with `try/except/else/finally`, raise errors, and define a custom exception.



```
PS C:\internship> & C:\Users\kamai\AppData\Local\Programs\Python\Python314\python.exe c:/internship/week1/06_exceptions.py
division attempted
Cannot divide by zero
division attempted
Both values must be numbers
division attempted
5.0 None None
Valid age: 25
Invalid: 'abc' is not a valid integer
Invalid: Age out of range
Order failed: laptop: requested 5, only 3 available
File error: No such file or directory
PS C:\internship>
```

Figure 6: Output of 06_exceptions.py.

Observations (Figure 6):

- `safe_divide` catches `ZeroDivisionError` and `TypeError`; the `finally` block prints “division attempted” on every call.
- `parse_age` accepts 25 and rejects “abc” (not an integer) and 150 (out of range).
- The custom `InsufficientStockError` reports that 5 laptops were requested but only 3 are available.
- Opening a missing file raises `FileNotFoundError`, handled without crashing the program.

9 Conclusion

All six scripts ran without errors and produced the expected outputs, which confirms a working understanding of Python fundamentals: data types and collections, control flow, functions, OOP, file I/O and exception handling. These skills are the basis for the next tasks (data cleaning and EDA with Pandas, and Scikit-learn models).